15.5 Configure Static Host Routes

Configure Static Host Routes
Host Routes

A host route is an IPv4 address with a 32-bit mask, or an IPv6 address with a 128-bit mask. The following shows the three ways a host route can be added to the routing table:

- Automatically installed when an IP address is configured on the router
- Configured as a static host route
- Host route automatically obtained through other methods (discussed in later courses)

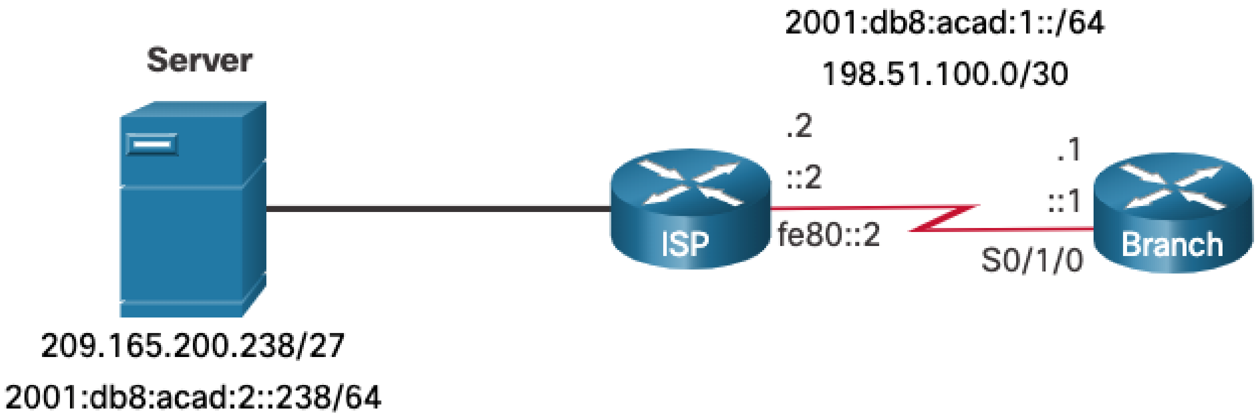
Automatically Installed Host Routes

- Cisco IOS automatically installs a host route, also known as a local host route, when an interface address is configured on the router. A host route allows for a more efficient process for packets that are directed to the router itself, rather than for packet forwarding.
- This is in addition to the connected route, designated with a **C** in the routing table for the network address of the interface.
- The local routes are marked with **L** in the output of the routing table.

Static Host Routes

A host route can be a manually configured static route to direct traffic to a specific destination device, such as the server shown in the figure.

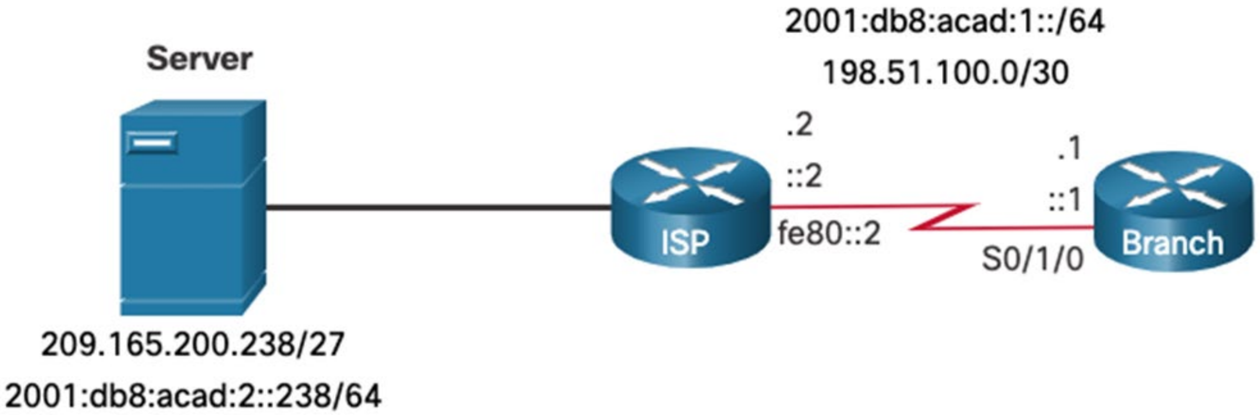
The static route uses a destination IP address and a **255.255.255.255** (/32) mask for IPv4 host routes, and **a /128** prefix length for IPv6 host routes.



Configure Static Host Routes

The example shows the IPv4 and IPv6 static host route configuration on the Branch router to access the server.

```
Branch(config)# ip route 209.165.200.238 255.255.255.255 198.51.100.2
Branch(config)# ipv6 route 2001:db8:acad:2::238/128 2001:db8:acad:1::2
Branch(config)# exit
Branch#
```



Verify Static Host Routes

A review of both the IPv4 and IPv6 route tables verifies that the routes are active.

```
Branch# show ip route | begin Gateway
Gateway of last resort is not set
      198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C        198.51.100.0/30 is directly connected, Serial0/1/0
L        198.51.100.1/32 is directly connected, Serial0/1/0
      209.165.200.0/32 is subnetted, 1 subnets
S        209.165.200.238 [1/0] via 198.51.100.2
Branch# show ipv6 route
(Output omitted)
C   2001:DB8:ACAD:1::/64 [0/0]
     via Serial0/1/0, directly connected
L   2001:DB8:ACAD:1::1/128 [0/0]
     via Serial0/1/0, receive
S   2001:DB8:ACAD:2::238/128 [1/0]
     via 2001:DB8:ACAD:1::2
Branch#
```

Configure IPv6 Static Host Route with Link-Local Next-Hop

For IPv6 static routes, the next-hop address can be the link-local address of the adjacent router. However, you must specify an interface type and an interface number when using a link-local address as the next hop, as shown in the example. First, the original IPv6 static host route is removed, then a fully specified route configured with the IPv6 address of the server and the IPv6 link-local address of the ISP router.

```
Branch(config)# no ipv6 route 2001:db8:acad:2::238/128 2001:db8:acad:1::2
Branch(config)# ipv6 route 2001:db8:acad:2::238/128 serial 0/1/0 fe80::2
Branch# show ipv6 route | begin ::
C   2001:DB8:ACAD:1::/64 [0/0]
     via Serial0/1/0, directly connected
L   2001:DB8:ACAD:1::1/128 [0/0]
     via Serial0/1/0, receive
S   2001:DB8:ACAD:2::238/128 [1/0]
     via FE80::2, Serial0/1/0
Branch#
```